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How Much Do Language Models Memorize?

GPT-Style Transformers Store Approximately 3.6 Bits Per Parameter—and Grokking Begins When Capacity Fills

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How much of a training dataset does a language model actually memorize? Prior answers conflated memorization with generalization. This paper separates them formally and measures each independently. The result is a clean constant: GPT-style transformers memorize approximately **3.6 bits per parameter**. A 7B-parameter model can hold roughly 25 GB of training content. The paper also explains grokking: it begins when the training dataset exceeds model capacity in bits, forcing a transition from memorization to generalization.

AT A GLANCE

Problem: Memorization and generalization cannot be disentangled by standard extraction attacks or train-test loss gaps, making capacity estimates unreliable.

Why it matters: Capacity sets an upper bound on privacy risk and copyright exposure; the grokking connection has implications for training regime selection.

Method: Train models on random bitstrings (zero generalization signal); measure information-theoretically how many bits the model stores per parameter.

Result: ≈ 3.6 bits per parameter across model sizes (100K–20M parameters); grokking onset is predicted by the capacity-dataset size threshold.

Evidence: Systematic sweep over model sizes and dataset sizes; mutual information estimates via compression proxy.

The Big Picture

Language models trained on internet text raise two practical concerns: privacy (memorization of personal data) and copyright (verbatim reproduction of protected works). Both depend on a prior question that has never been answered cleanly: how much can a language model actually memorize, expressed as an absolute information-theoretic

quantity?

This paper provides that answer. It introduces a clean experimental design that eliminates generalization entirely, a formal decomposition of model information into memorization and generalization components, and a measured constant—3.6 bits/parameter—that holds across model scales.

Why Existing Approaches Fall Short

Extraction attacks (membership inference, verbatim reproduction) measure whether a training string can be recovered, not how much information the model contains about it. A model that has generalized strongly might reproduce a training string without memorizing it specifically.

Train-test loss gaps measure the total effect of training data on model outputs, which conflates memorization (dataset-specific information) with generalization (distributional information). A model trained on more data can show a smaller loss gap while containing *more* total memorized information, confusing the measurement.

Both approaches systematically underestimate capacity because they cannot access the true mutual information between model weights and training data.

Core Mechanism

The paper eliminates generalization by design: training sets consist of uniformly random bitstrings. No two bitstrings share any statistical structure, so any information a model acquires about training data is pure memorization—there is no distribution to generalize about.

Models are trained to completion on these datasets, and the number of bitstrings memorized (measured by next-bit prediction accuracy approaching $\log 2$) is recorded as a function of model size and dataset size.

The ratio (bits memorized) / (model parameters) converges to 3.6 across all tested architectures and sizes, establishing this as a model-class constant for GPT-style transformers.

Technical Formulation

Let M be the model parameters, $Z = (Z_1, \dots, Z_n)$ the training dataset, and $\mathcal{D}$ the true data-generating distribution. The decomposition of memorization is:

$$I(M; Z_i) = \underbrace{I(M; Z_i | \mathcal{D})}_{\text{unintended memorization}} + \underbrace{I(M; \mathcal{D})}_{\text{generalization}}.$$

On random bitstrings, $\mathcal{D} = \text{Uniform}(\{0, 1\}^\ell)$ and $I(M; \mathcal{D}) = 0$ by independence, so:

$$I(M; Z_i) = I(M; Z_i | \mathcal{D}).$$

Total memorization capacity is:

$$C = \sum_{i=1}^n I(M; Z_i | \mathcal{D}).$$

The empirical estimate uses:

$$C \approx \sum_{i=1}^n \ell_i - H(Z_i | M),$$

where $H(Z_i | M)$ is estimated from the model’s per-bit cross-entropy on held-out bitstrings (not in training set) minus the per-bit cross-entropy on memorized bitstrings. The gap directly quantifies memorized information.

Learning or Inference Procedure

1. Generate datasets $\mathcal{D}_n$ of n random uniform bitstrings of length $\ell = 64$ for $n \in \{100, 1000, 10000, 100000\}$.
2. Train GPT-style transformers of varying sizes ($P \in \{100\text{K}, 500\text{K}, 2\text{M}, 10\text{M}, 20\text{M}\}$ parameters) on each dataset to convergence.
3. Measure $H(Z_i | M)$ by next-bit cross-entropy on training strings; compare to held-out bitstrings to compute $I(M; Z_i)$.
4. Plot total bits memorized vs. P ; estimate slope = capacity per parameter.
5. Identify the dataset size n^* at which loss curve inflects (grokking onset) and verify $n^* \approx C/\ell$ (bits per string).

What the Guarantee Says

The 3.6 bits/parameter figure is empirical but consistent across: (a) model sizes spanning two orders of magnitude, (b) three different transformer architectures (GPT-2 style, LLaMA-style, BERT-style masked), (c) bitstring lengths $\ell \in \{32, 64, 128\}$.

The theoretical lower bound from data-processing inequality is 1 bit/parameter (any lossless compression of P parameters into P floats); the empirical constant of 3.6 reflects the overhead of the transformer’s implicit prior.

Experimental Findings

Model Size	Params	Cap. (Mb)	Bits/Param
Tiny	100K	0.36 Mb	3.6
Small	2M	7.2 Mb	3.6
Med.	10M	36 Mb	3.6
Large	20M	72 Mb	3.6

Grokking onset: for all model sizes, sudden generalization (loss drop on structured held-out tasks) occurs when $n \approx C/\ell$ (within $\pm 8\%$ across runs), confirming the capacity-threshold theory.

Ablations and Interpretation

Architecture variants: Switching from GPT-2 attention to grouped-query attention (LLaMA-style) changes the constant from 3.6 to 3.7, indicating architecture affects efficiency marginally but not the order of magnitude.

Precision: FP16 training reduces effective capacity to ≈ 3.1 bits/parameter; BF16 yields 3.4; FP32 yields 3.6. Lower-precision arithmetic has a small but consistent capacity cost.

Privacy implication: A 7B-parameter model (e.g. Llama-3-7B) has capacity $7 \times 10^9 \times 3.6 \approx 25$ Gb. A typical book is 1–2 Mb; this model could in principle memorize $\approx 10,000$ to 25,000 books verbatim, absent any regularization or curriculum.

Grokking implication: Training beyond n^* is not wasted; it triggers the memorization-to-generalization phase transition and can reduce unintended memorization while improving OOD performance.

Reference

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